distribution of matter and energy throughout space. In the mechanism Guth proposed, this conversion would happen through the collision of bubble walls, but calculations—carried out by Guth and Erick Weinberg of Columbia University, and also by Stephen Hawking, Ian Moss, and John Steward of Cambridge University—revealed that the resulting distribution of matter and energy was *not* uniform. Thus, Guth's original inflationary model ran into significant problems of detail.

The insights of Linde and of Albrecht and Steinhardt—now called *new inflation*— fixed these vexing problems. By changing the shape of the potential energy bowl to that in Figure 10.2, these researchers realized, the inflaton could relax to its zero energy value by "rolling" down the energy hill to the valley, a gradual and graceful process that had no need for the quantum jump of the original proposal. And, as their calculations showed, this somewhat more gradual rolling down the hill sufficiently prolonged the inflationary burst of space so that one single bubble easily grew large enough to encompass the entire observable universe. Thus, in this approach, there is no need to worry about coalescing bubbles. What was of equal importance, rather than converting the inflaton field's energy to that of ordinary particles and radiation through bubble collisions, in the new approach the inflaton gradually accomplished this energy conversion uniformly throughout space by a process akin to friction: as the field rolled down the energy hill —uniformly throughout space—it gave up its energy by "rubbing against" (interacting with) more familiar fields for particles and radiation. New inflation thus retained all the successes of Guth's approach, but patched up the significant problem it had encountered.

About a year after the important progress offered by new inflation, Andrei Linde had another breakthrough. For new inflation to occur successfully, a number of key elements must all fall into place: the potential energy bowl must have the right shape, the inflaton field's value must begin high up on the bowl (and, somewhat more technically, the inflaton field's value must itself be uniform over a sufficiently large spatial expanse). While it's possible for the universe to achieve such conditions, Linde found a way to generate an inflationary burst in a simpler, far less contrived setting. Linde realized that even with a simple potential energy bowl, such as that in Figure 9.1a, and even without finely arranging the inflaton field's initial value, inflation could still naturally take place. The idea is this. Imagine that in the very early universe, things were "chaotic"—for example, imagine that there was an inflaton field whose value randomly bounced around from one number to another. At some locations in space its value might have been small, at other locations its value might have been medium, and at yet other locations in space its value might have been high. Now, nothing particularly noteworthy would have happened in regions where the field value was small or medium. But Linde realized that something fantastically interesting would have taken place in regions where the inflaton field happened to have attained a high value (even if the region were tiny, a mere 10^{-33} centimeters across). When the inflaton field's value is high—when it is high up on the energy bowl in Figure 9.1a— a kind of cosmic friction sets in: the field's value tries to roll down the hill to lower potential energy, but its high value contributes to a resistive drag force, and so it rolls very slowly. Thus, the inflaton field's value would have been nearly constant and (much like an inflaton on the top of the potential energy hill in new inflation) would have contributed a nearly constant energy and a nearly constant negative pressure. As we are now very familiar, these are the conditions required to drive a burst of inflationary expansion. Thus, without invoking a particularly special potential energy bowl, and without setting up the inflaton field in a special configuration, the chaotic environment of the early universe could have naturally given rise to inflationary expansion. Not